Coupling a multi-species point model to a continuum biofilm growth model: verification in three codes and the effect of the composition on growth and stress

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Abstract

Continuum models of biofilm growth predict where biomass forms and the stresses it causes, but treat the biofilm as one material. Multi-species point models describe the composition, the living fraction and the response to an antibiotic, but have no space. This work couples the multi-species point model of Klempt et al. (2026), used as published, to the Hamilton-principle growth model of Klempt et al. (2024) at every integration point, without changing the equations of either model. The point model receives the local nutrient and antibiotic from the fields, and its living volume fraction $\sum_i \bar{\phi}_i$ sets the consumption $g\,\phi_{3D}(\sum_i \bar{\phi}_i)\,c$ of both. The structure of the coupled equations fixes what the composition can change: it reaches growth and stress only through the nutrient and the front speed of the growth equation, because the growth law counts all biomass, so the growth is bounded by $k_\alpha T^*$ and the front speed by a factor of two. The quasi-steady nutrient depends on the composition through one product and is depleted over the half edge of the domain (Thiele modulus 1). In the point model the antibiotic acts through a balance whose decay rate grows as the inverse square of the species' volume fraction, so a dense species is removed slowly and a thinly present one at once. Examples with two species and an antibiotic, computed in a finite-volume code and in user elements of ANSYS and Abaqus and verified by a manufactured solution and grid convergence indices, confirm this: the antibiotic removes most of the living biomass of the seed (0.97 to 0.15 when the species start apart, 0.96 to 0.04, ten times faster, when mixed), while the growth changes by about 1 % and the converged seed stress within its grid uncertainty; the local nutrient decides where the antibiotic acts; and the state that newly reached material receives, which neither model fixes, decides in a maze whether the biofilm recovers. The same holds around a human molar crown. A second-order treatment of the front term is needed for the stress.

Keywords: biofilm growth; multi-species model; antibiotic; extended Hamilton principle; code verification; grid convergence index; growth-induced stress

1 Introduction

Oral biofilms on teeth and implants are communities of several bacterial species. Their composition changes over days: early colonisers prepare the surface for later ones, and pathogenic species appear only when the community allows it (Heine et al., 2025). Treatments such as antibiotics act on the composition, while the mechanical load on the surface comes from the amount and the growth of the biomass. A model that predicts the stresses for a given treatment therefore has to describe both: where the biofilm grows, and what it is made of.

Two kinds of model exist for this. Continuum growth models describe the amount of biomass in space. Wanner and Gujer (1986) introduced the one-dimensional multispecies biofilm, Eberl

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et al. (2001) a density-dependent diffusion model of the spreading. Feng et al. (2021) model a two-species oral biofilm in which the spreading follows the summed growth of the species, and Soleimani et al. (2023) the co-aggregation of two species in a user element. Klempt et al. (2024) derive the growth of a biofilm from the extended Hamilton principle (Junker and Balzani, 2021), with a phase field ϕ_{3D}, a nutrient c and the growth variable α of a multiplicative split $\mathbf{F} = \mathbf{F}_e\mathbf{F}_g$, and compute the resulting stresses; their model has one biomass and no composition. Point models describe the composition of a community at one point. Klempt et al. (2026) derive a multi-species model from the same variational principle, with volume fractions ϕ_i and living fractions ψ_i of N species, an interaction matrix $\mathbf{A}$ and the response of each species to an antibiotic; Fritsch et al. (2025) calibrate its parameters by Bayesian updating. These models have no space: the nutrient c^* and the antibiotic α^* are given inputs.

The point model needs local values of c^* and α^*, which a growth field can supply. The growth field needs to know how much of the biomass is alive, because only living biomass consumes nutrient. Connecting the two at every material point is the natural step, but it raises questions that neither paper answers: which variables are passed, whether either model has to be changed, which new assumptions are needed (for example the composition of biofilm that a point reaches later), and whether the composition reaches the growth and the stress at all.

This work couples the point model of Klempt et al. (2026) to the growth model of Klempt et al. (2024) without changing either set of equations. The contribution is threefold.

1. The point model runs at every integration point as published. It receives $c^* = c_0^* c$ and $\alpha^* = \alpha_0^* a$ from the nutrient and antibiotic fields and returns the living volume fraction $\sum_i \bar{\phi}_i$, which sets the consumption $g\,\phi_{3D}(\sum_i \bar{\phi}_i)\,c$ in the field equations.

2. What the composition can change follows from the structure of the coupled equations: the path from the living fraction to the stress, a bound on the growth, the quasi-steady nutrient with its Thiele modulus, the balance by which the antibiotic acts in the point model and its dependence on the density of a species, and the meaning of the time link. These statements are derived from the published equations and then read against the examples.

3. Examples with two species and an antibiotic, with the synthetic data of Fritsch et al. (2025) and the constants of Klempt et al. (2024, 2026), computed in a finite-volume code and in user elements of ANSYS and Abaqus and verified by a manufactured solution and grid convergence indices, show what the composition changes and what it does not, which assumption beyond the two papers matters, and the coupling around a human molar crown.

Section 2 states the models, the coupling and what its structure implies, Section 3 the implementations and their verification in short (details in the appendix), Section 4 the examples, Section 5 the limits of the coupling, and Section 6 concludes.

2 Models and coupling

2.1 Growth field

The biofilm is described by the phase field $\phi_{3D} \in [0, 1]$, the nutrient c and the growth variable α of Klempt et al. (2024), with $\mathbf{F} = \mathbf{F}_e\mathbf{F}_g$, $\mathbf{F}_g = \alpha\mathbf{I}$ and $\alpha(0) = 1$, so that the growth is $\alpha - 1$. After division by the viscosities, their Eqs. 34 to 36 read

$$\dot{\phi}_{3D} = \beta\,\nabla^2\phi_{3D} + k_\alpha\,\alpha - r\,\frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla\phi_{3D}, \qquad \mathbf{n}_c = \frac{\nabla c}{|\nabla c|}, \tag{1}$$

$$\dot{c} = d\,\nabla^2 c - g\,\phi_{3D}\Big(\textstyle\sum_i \bar{\phi}_i\Big)\,c, \tag{2}$$

$$\dot{\alpha} = k_\alpha\,\phi_{3D}, \tag{3}$$

with zero flux on the boundary except where the nutrient is held. The last term of Eq. (1) moves the edge of the biofilm up the nutrient gradient with the speed $r\,c/(k+c)$; we call it the front term. Two details differ from the equations as printed, both confirmed by the authors: the consumption of Eq. (2) is $g\,\phi\,c$ (the printed $g\,\phi$ is a misprint), and the front direction is regularised as $\nabla c/(|\nabla c|^2+\varepsilon^2)^{1/2}$ with $\varepsilon^2 = 10^{-6}$ ($\varepsilon^2 = 10^{-8}$ changes the seed stress by 0.1 %). The factor $\sum_i \bar{\phi}_i$ in Eq. (2) is the coupling term of this work (Section 2.3); in Klempt et al. (2024) the whole biofilm consumes. The mechanical term $\phi\,\mu\,(\mathbf{I}:\mathbf{C}_e - 3)$ of their Eq. 30, which they neglect because μ is small against the other parameters, is neglected here as well, so the mechanics is coupled one way. The elastic response is neo-Hookean with the stiffness $E(\phi_{3D}) = E\,(\phi_{3D}^2 + f)$, $E = 10$ Pa, $\nu = 0.49$ and a void floor $f = 10^{-3}$ (an assumption that keeps the empty cells solvable; $f \to 0$ changes the seed stress by 0.13 %).

2.2 Point model

At a material point the multi-species model of Klempt et al. (2026) describes the volume fractions ϕ_i and the living fractions ψ_i of N species and the void ϕ_0, with $\sum_i \phi_i + \phi_0 = 1$. It follows from the extended Hamilton principle with the free energy $\Psi = -\frac{1}{2} c^*\,\bar{\boldsymbol{\phi}}^{\mathrm{T}}\mathbf{A}\,\bar{\boldsymbol{\phi}}$, $\bar{\phi}_i = \phi_i\psi_i$, a dissipation with the viscosities η_i, $\eta_{\phi,i}$ and the constraint enforced by the multiplier γ:

$$\begin{aligned} (\eta_{\phi,i} + \eta_i\psi_i^2)\,\dot{\phi}_i + \eta_i\phi_i\psi_i\,\dot{\psi}_i &= c^*\psi_i \textstyle\sum_j A_{ij}\bar{\phi}_j - \gamma - (\text{barrier}),\\ \eta_i\,(\phi_i\psi_i\,\dot{\phi}_i + \phi_i^2\,\dot{\psi}_i) &= c^*\phi_i \textstyle\sum_j A_{ij}\bar{\phi}_j - \alpha^* b_i\,\psi_i - (\text{barrier}), \end{aligned} \tag{4}$$

where A_{ij} are the interactions, b_i the sensitivity of species i to an antibiotic of concentration α^*, c^* the nutrient, and the barrier terms keep ϕ_i, ψ_i in $(0,1)$. The antibiotic lowers ψ_i through the term $\alpha^* b_i \psi_i$, against the nutrient term $c^*\phi_i \sum_j A_{ij}\bar{\phi}_j$ (Section 2.4). Our implementation reproduces the fourteen numerical examples of the paper (two printed details read differently: Eq. 17 as printed carries a $+\gamma$ that the figures do not show, and the four-species interaction matrix of Eq. 19 reproduces the figures without its prefactor $\frac{1}{2}$) and the figures of Fritsch et al. (2025) within 0.013. The model is used as published, with its own constraint and its implicit time integration (steps of 10^{-4} in its own time unit, Newton's method). For the examples the synthetic two-species data of Fritsch et al. (2025), case I, are used with their true parameters: $\mathbf{A} = \left(\begin{smallmatrix} 1 & 0.1 \\ 0.1 & 1 \end{smallmatrix}\right)$, $b = (1, 2)$, $\eta = (1, 2)$, $c_0^* = 100$.

2.3 Coupling

The two models are connected at every integration point without changing either (Figure 1, Algorithm 1).

1. *Fields to point model.* The point model sees $c^* = c_0^*\,c$ and $\alpha^* = \alpha_0^*\,a$, with a the antibiotic, transported by a second field with the diffusivity and uptake of the nutrient, $\dot{a} = d\,\nabla^2 a - g\,\phi_{3D}(\sum_i \bar{\phi}_i)\,a$. Between the increments t_n and t_{n+1} of the field both are interpolated linearly in time over the inner steps of the point model,

$$c^*(t) = c_0^*\Big[\tfrac{t_{n+1}-t}{\Delta t}\,c_n + \tfrac{t-t_n}{\Delta t}\,c_{n+1}\Big], \qquad t_n \le t \le t_{n+1}. \tag{5}$$

   Klempt et al. (2026) already use a time-dependent c^* (their four-species case 3), so the point model accepts such an input without change.

2. *Point model to fields.* Only the living biomass consumes: the uptake of nutrient and antibiotic is $g\,\phi_{3D}(\sum_i \bar{\phi}_i)$, Eq. (2). The point model normalises its volume, so $\bar{\phi}_i \in [0,1]$ is the living volume fraction of species i at the point and $\phi_{3D}\sum_i \bar{\phi}_i \le \phi_{3D}$ the living volume fraction in the field; the void ϕ_0 has no living fraction and is not part of the sum. A dead biofilm consumes nothing. The growth law (3) is unchanged: α measures the biomass produced, alive or dead.

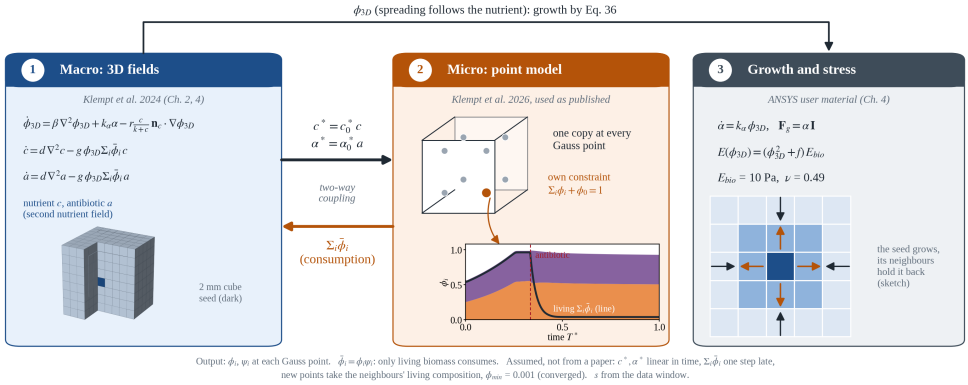


Figure 1: The coupling: the fields ϕ_{3D}, c and a of the growth model, the point model at every integration point with the inputs $c^* = c_0^* c$, $\alpha^* = \alpha_0^* a$ and the output $\sum_i \bar{\phi}_i$, and the mechanics with $\mathbf{F}_g = \alpha \mathbf{I}$.

Algorithm 1 One increment $t_n \to t_{n+1}$ of the coupled run (user-element form)

Require: at every integration point α_n, the point-model state $(\phi_i, \psi_i, \gamma)_n$ and the pair (c, a) seen in the previous increment; the fields ϕ_{3D}, c, a of t_n
Ensure: displacements, stresses and state of t_{n+1}

repeat ▷ Newton iteration of the equilibrium
  for all integration points **do**
    $\alpha_{n+1} \leftarrow \alpha_n + k_\alpha \, \phi_{3D} \, \Delta t$ ▷ Eq. (3)
    $\mathbf{F}_e \leftarrow \mathbf{F} \, \mathbf{F}_g^{-1}$, $\mathbf{F}_g = \alpha_{n+1} \mathbf{I}$; stress and tangent from $W(\mathbf{F}_e)$ and $E(\phi_{3D})$
    if $\phi_{3D} \geq \phi_{\min}$ or the point model has started **then**
      if not yet advanced in this increment **then**
        (c_n, c_{n+1}), (a_n, a_{n+1}) from the previous and the current value at the point
        n_{sub} steps of $s\Delta t / n_{\text{sub}}$ of Eq. (4) with c^*, α^* from Eq. (5)
        store $(\phi_i, \psi_i, \gamma)_{n+1}$ and $\sum_i \bar{\phi}_i$
      end if
    end if
  end for
until the residual meets the convergence criterion
advance ϕ_{3D} (Eq. (1)), c and a (Eq. (2) with $\sum_i \bar{\phi}_i$)

3. *Time.* Klempt et al. (2024) normalise time with a reference time chosen for the problem at hand. We take $T^* = 1$ as the duration covered by the point model's data, so that both models run on the same time: the point model advances $s\,\Delta T^*$ per increment, with s the length of that window in its own units, not a fitted value ($s = 0.1$ for case I, 1000 steps of 10^{-4}).

4. *Start of the point model.* It starts at a point once ϕ_{3D} there reaches $\phi_{\min} = 0.001$. Where a seed is placed it starts from the seed composition. A point that the spreading biofilm reaches later takes the living composition of its colonised neighbours (their amounts divided in the ratio of $\bar{\phi}_i$) and starts alive ($\psi_i = 0.999$, $\gamma = 0$), because the biofilm spreads through the growth of its living cells. The papers do not fix this rule; it is an assumption, and the alternative of starting from the seed composition is compared in Section 4.4.

What this coupling can and cannot change follows from the structure of the equations and is stated next.

2.4 What the structure of the coupled equations implies

Four properties of the coupled system follow from Eqs. (1) to (4) alone and are used to read the examples.

The path from the composition to the stress. The composition enters the field equations only through the factor $\sum_i \bar{\phi}_i$ in the consumption of Eq. (2). The nutrient enters Eq. (1) only in the front term, through the speed $r\,c/(k+c)$ and the direction $\mathbf{n}_c$, and it does not enter the growth law (3) at all. The composition therefore reaches ϕ_{3D}, α and the stress along one path only,

$$\textstyle\sum_i \bar{\phi}_i \xrightarrow{(2)} c \xrightarrow{\text{front term of (1)}} \phi_{3D} \xrightarrow{(3)} \alpha \longrightarrow \boldsymbol{\sigma}, \tag{6}$$

and it has no effect wherever that path is cut: in the interior of a biofilm, where $\nabla\phi_{3D} = \mathbf{0}$, and wherever the nutrient is uniform, where $\nabla c = \mathbf{0}$ and the front term vanishes. Two bounds follow. Since $0 \le \phi_{3D} \le 1$, integrating Eq. (3) gives $\alpha(T) - 1 = k_\alpha \int_0^T \phi_{3D}\,dt \le k_\alpha T$; in the seed of the examples $\alpha - 1$ reaches 0.91 of this bound at $T^* = 1$, so no change of the composition can raise the seed growth by more than about 10 %, and it can lower it only through the edge of the seed, where $\phi_{3D} < 1$ enters the mean. And the front speed $r\,c/(k+c)$ with $k = 1$ and $0 \le c \le 1$ lies between 0 and $r/2$, so the composition changes the speed of the edge by at most a factor of two through the nutrient, and in practice by the ratio $(c/(1+c))$ at the two nutrient levels (by 37 % between the seed means 0.54 and 0.91 of Section 4.2). The stress is set by the jump of α across the edge of the seed, so the composition acts on the stress through the position and sharpness of that edge, not through the amount of growth. Both bounds are properties of the published equations: a growth law in which the rate counts the living biomass, $\dot{\alpha} = k_\alpha \phi_{3D} \sum_i \bar{\phi}_i$, or a front speed multiplied by $\sum_i \bar{\phi}_i$, would open a second path, and both are changes of the model (Section 5).

The nutrient is quasi-steady and depends on the composition through one product. With the constants of Table 1 the nutrient adjusts within about $4 \times 10^{-8}\,T^*$, so Eq. (2) reduces to the linear problem

$$\nabla^2 c = \frac{\Lambda^2}{\ell^2}\,\kappa\,c, \qquad \kappa = \phi_{3D} \textstyle\sum_i \bar{\phi}_i, \qquad \Lambda^2 = \frac{g\ell^2}{d} = 1, \tag{7}$$

with $\ell = 1$ mm the half edge of the cube and Λ the Thiele modulus of the conversion. Three consequences follow. The nutrient is proportional to its held value, so the examples do not depend on the level of the supply. It depends on the composition only through the living volume fraction κ of the field, not on which species is alive. And it is depleted over the length $\lambda = \ell/(\Lambda\sqrt{\kappa})$, which is 1 mm, the half edge of the cube, for fully living biofilm and 2.6 mm when $\sum_i \bar{\phi}_i = 0.15$: with $\Lambda = 1$ the nutrient is never exhausted inside the biofilm, it decreases by a factor of order one across the cube. A slab of living biofilm with uniform κ between the supply face and the opposite wall gives $c = \cosh((\ell - y)/\lambda)/\cosh(2\ell/\lambda)$ and at the centre 0.41 for $\kappa = 1$ and 0.82 for $\kappa = 0.15$; the seed means of Section 4.2 (0.54 and 0.91) lie above these slab values because the biofilm outside the seed has $\phi_{3D} < 1$ and does not fill the cube. The antibiotic field obeys the same equation with the same κ, so it is depleted over the same length; where living biofilm shields the nutrient it shields the antibiotic as well.

The antibiotic acts through a balance that depends on the density of the species. The living fraction of species i evolves by the second line of Eq. (4), Eq. 17 of Klempt et al. (2026). At fixed ϕ_i and away from the barriers,

$$\eta_i\,\phi_i^2\,\dot{\psi}_i = c^*\phi_i \textstyle\sum_j A_{ij}\phi_j\psi_j - \alpha^* b_i\,\psi_i, \tag{8}$$

so ψ_i rises where the nutrient term exceeds the antibiotic term and falls where it does not. For a species that dominates its point ($A_{ii}\phi_i^2\psi_i \gg \sum_{j\neq i} A_{ij}\phi_j\psi_j$) the balance is $c^* A_{ii}\phi_i^2$ against $\alpha^* b_i$, and the relative rate is

$$\frac{\dot{\psi}_i}{\psi_i} = \frac{c^* A_{ii}}{\eta_i} - \frac{\alpha^* b_i}{\eta_i\,\phi_i^2}. \tag{9}$$

The nutrient term gives a rate that does not depend on how much of the species is present, the antibiotic term a decay rate that grows as $1/\phi_i^2$: the antibiotic kills a species that fills its point slowly and the same species present with a small volume fraction almost at once. The rate changes sign on the line $c^* A_{ii}\phi_i^2 = \alpha^* b_i$, but ψ_i decays exponentially rather than reaching zero, and ϕ_i evolves at the same time by the first line of Eq. (4), so the equations give a balance with a graded outcome, not a sharp boundary between living and dead regions. In the examples $c^* = c_0^* c$ and $\alpha^* = \alpha_0^* a$ with $c_0^* = \alpha_0^*$, so the balance for species 1 ($b_1 = 1$) reads $c\,\phi_1^2$ against a: it is approached only where the nutrient is as high as the antibiotic and the species fills its point, next to the supply face, and everywhere else the species declines at a rate set by $a/\phi_1^2 - c$. Where the nutrient is high the decline is slow, where it is low it is fast; this is the spatial pattern of Section 4.2. The same expression says that a mixed seed, in which each species holds about half the volume, loses its living fraction much faster than a seed of one species, and that the state given to newly reached material decides its fate under the antibiotic (Section 4.4).

The time link is not a parameter of the model. Both lines of Eq. (4) are linear in the rates, and every rate carries the viscosity of its species. Running the point model on the time T^* of the field with $t = s\,T^*$ therefore equals running it on its own time with the viscosities η_i/s and $\eta_{\phi,i}/s$ (the barrier penalty $K_{p,i} = 10^{-4}\eta_i$ scales with them). With $T^* = 1$ taken as the duration of the data the point model was calibrated on, s is fixed by the data and the viscosities keep the values of the calibration; no new constant enters.

2.5 Parameters

All constants are those of the papers (Table 1): Table 2 of Klempt et al. (2024), converted from their 20 μm cube to the 2 mm cube of the examples, and the antibiotic of case 4 of Klempt et al. (2026), $\alpha_0^* = 100$ from $T^* = 1/3$. With $d = 10^8$ mm$^2/T^*$ the fields adjust within about $4 \times 10^{-8}\,T^*$, so the fields are quasi-steady; keeping the time derivative of Eq. (2) changes the seed values by less than 0.005. A diffusion slower by eight orders of magnitude ($d = 1$ mm$^2/T^*$) is only a sensitivity: the antibiotic then reaches the seed more slowly and kills markedly less of it. Values not given by the papers are marked as assumptions: the length and time scale of the conversion, $\phi_{\min}$ (results change by less than 1 % below 0.001), the start rule of new points, the void floor f and the regularisation ε.

3 Implementations and verification

The coupled model is solved in three codes that share the equations, the parameters and the compiled point-model routine, but not the discretisation: a finite-volume code (7-point Laplacian, second-order upwind front term with the minmod limiter, RK4 in time, hexahedral mechanics with selective reduced integration), the neighboured-element user element of the original authors' group in ANSYS (Blaszczyk et al., 2022; Rudolf et al., 2025), in which the Laplacian and the gradients are fitted on the Gauss points from 30 neighbours by weighted least squares and the fields are updated explicitly, and a Galerkin user element in Abaqus (8-node hexahedra, SUPG (Brooks and Hughes, 1982) for the front term, $\bar{\mathbf{B}}$ mechanics, implicit Euler with Newton's method). Three numerical details of the neighboured element were corrected before use, without changing its equations (Appendix A.1). Differences between the codes therefore come from the fields and the mechanics only.

Table 1: Constants of the examples. Conversion: Table 2 of Klempt et al. (2024) on the 2 mm cube with the half edge $\ell = 1$ mm, so that $g\ell^2/d = 1$.

quantity	symbol	value	source
diffusion of ϕ_{3D}	β	0.02 mm$^2/T^*$	Klempt et al. (2024), converted (assumption)
front speed, half saturation	r, k	10 mm/T^*, 1	Klempt et al. (2024), converted
growth rate	k_α	$10^{-3}\,T^{*-1}$	Klempt et al. (2024)
nutrient diffusion, uptake	d, g	10^8 mm$^2/T^*$, $10^8\,T^{*-1}$	Klempt et al. (2024), converted
antibiotic diffusion, uptake	d_a, g_a	$= d$, $= g$	assumption
stiffness, Poisson ratio	E, ν	10 Pa, 0.49	Klempt et al. (2024)
void floor	f	10^{-3}	assumption
point model	$\mathbf{A}$, b, η, c_0^*	case I	Fritsch et al. (2025)
antibiotic level, onset	α_0^*, T_{ab}^*	100, 1/3	Klempt et al. (2026), case 4
time link	s	0.1	from the data of case I
start threshold	$\phi_{\min}$	0.001	assumption (converged)
front regularisation	ε^2	10^{-6}	assumption

The verification follows the usual steps (Roache, 2002; Celik et al., 2008); Appendix B gives the details. With uniform fields every code reproduces the point model alone to round-off (2×10^{-14} in the finite-volume code, 1.8×10^{-12} in the element), so the hand-over of $\sum_i \bar{\phi}_i$ to the consumption is exact. With a manufactured solution the finite-volume and the Galerkin code converge at second order in L_2 for ϕ_{3D} and c (observed orders 1.7 to 2.05; the SUPG form of ϕ_{3D} at about 1.5), and the analytic tangent of the Galerkin element agrees with central differences to 5×10^{-11}; the neighboured element does not converge next to the walls (Appendix A.1). The mechanics of two codes agree to 1 % on the same growth field, so differences in the stress between the codes come from the growth field.

One numerical property of the model sets the accuracy of the stress and is therefore stated here. The front term of Eq. (1) is an advection with the speed $r\,c/(k + c)$ along $\mathbf{n}_c$ against the small diffusion β: the cell Péclet number $rh/(2\beta)$ is about 31 on a 16^3 grid of the 2 mm cube, so a central discretisation oscillates and needs stabilisation. A constant added diffusion $rh/4$, which keeps the element as delivered stable, smears the edge of the seed over a few cells; the seed stress, which is set by the jump of α across that edge, follows the smearing and is about a quarter of the converged value on 16^3 and still rising on 96^3. The growth variable converges with every treatment. The three second-order treatments (upwind with the minmod limiter, SUPG, upwind-biased least-squares gradients) give the same seed stress within a few per cent from 32^3 on, and the three codes agree within their grid-convergence bands (Appendix B.2). All results below use a second-order treatment.

The examples were then run on 32^3, 64^3 and 128^3 cells with the outer step 0.0025 (the step study is in Appendix B.3) and the three-grid index of Celik et al. (2008) with the safety factor 1.25 (Figure 2, Table 3). The growth converges with a GCI of 0.7 to 3.3 % in all examples, the seed stress with 0.7 % without and 8.3 % with the antibiotic when the species start apart. When they start mixed the stress converges slowly (observed order about 0.75, GCI 26 to 29 %), because the front carries intermediate values of ϕ_{3D} at a nearly discontinuous edge and the largest ϕ_{3D} in the seed still rises with the grid (0.60, 0.69, 0.73); with the start fractions of Fritsch et al. (2025) the 128^3 grid is not yet in the asymptotic range. No run on 32^3 to 128^3 left $[0, 1]$ by more than 2×10^{-4}.

TODO: Fourth grid (192^3, step limit of Appendix B.3) for the mixed example with the start fractions of Fritsch et al.; add to Table 3 and Figure 2 when available.

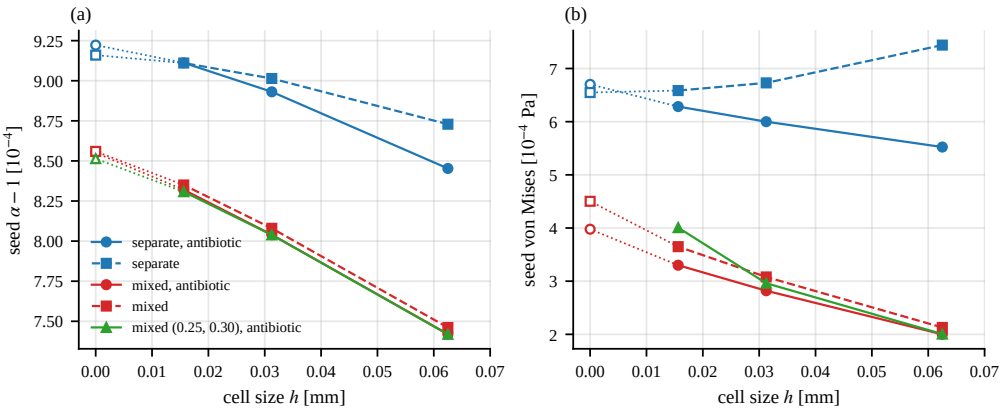


Figure 2: Grid convergence of the two-species examples (finite-volume code, outer step 0.0025; Table 3): seed means of (a) $\alpha - 1$ and (b) the von Mises stress at $T^* = 1$ against the grid. Dashed: Richardson extrapolation where the three grids are in the asymptotic range.

4 Results

4.1 Set-up

The examples use a 2 mm cube with a cubic seed of $\phi_{3D} = 1$ at its centre, the nutrient held at $c = 1$ on the face $y = -1$ mm and the antibiotic held at $a = 1$ on all outer faces from $T^* = 1/3$. In the separate example species A fills the seed and species B four small blocks along the edges of the cube; in the mixed example both start in the seed, in equal amounts ($\phi_1 = \phi_2 = 0.275$) or in the proportions of Fritsch et al. (2025) (0.25, 0.30). The front term is on in every run. The values below are seed means at $T^* = 1$ from the Python code (TVD, newly reached points inherit the living composition of their neighbours), which reaches the finest grids; the element gives the same composition within its grid error and its growth within 4 % after extrapolation. The stress is solved on 16^3 hexahedra with the element's integration; solving the same 64^3 field on 32^3 and 64^3 elements changes the seed von Mises stress by 0.2 %.

4.2 Composition and living biomass

Table 2 gives the seed means and Figure 3 the fields of the separate and the mixed example with antibiotic. Without antibiotic the living volume fraction of the seed stays near 0.96; the antibiotic lowers it to 0.15 when the species start apart and to 0.04 when they start mixed. Species B, the more sensitive one ($b_2 = 2$), is removed first. The nutrient field decides where the antibiotic acts, as Eq. (9) says: species A declines at the rate $(a/\phi_1^2 - c)\, c_0^* s/\eta_1$ per unit of T^* ($\eta_1 = 1$), so where the nutrient is high it survives, and where it is low it dies as well. The seed values give the size of the rates: in the separate example the seed holds $\phi_1 = 0.91$ with $c = 0.91$ and $a = 0.98$ at $T^* = 1$, a decline of 2.7 per unit of T^*, consistent with the fall of ψ_1 from 0.98 to 0.16 between the onset and $T^* = 1$; in the mixed seed species A holds only $\phi_1 = 0.57$, the decline is 21 per unit of T^*, and its living fraction is gone within $0.2\,T^*$ of the onset. The difference between 0.15 and 0.04 is therefore a property of the point model's balance, not of the field: the same species at the same nutrient and antibiotic dies ten times faster when it shares its point. When the biomass dies the nutrient is no longer consumed and rises in the seed, from 0.54 to 0.91 in the separate example, as the depletion length of Eq. (7) grows from 1 to 2.6 mm; the antibiotic, which obeys the same equation, reaches 0.98 in the seed. Living volume fraction, shares and nutrient change by less than 3 % between the grids (32^3 to 128^3) and the outer steps (0.02 to 0.0025) used.

Table 2: Seed means at $T^* = 1$, Python, 64^3 cells, outer step 0.025; converged growth and stress in Table 3.

example	$\sum_i \bar{\phi}_i$	$\phi_1/(\phi_1+\phi_2)$	c	$\alpha - 1$	von Mises [Pa]
separate	0.965	0.978	0.536	9.06×10^{-4}	6.56×10^{-4}
separate, antibiotic	0.152	0.951	0.911	8.95×10^{-4}	5.68×10^{-4}
mixed, equal amounts	0.961	0.859	0.608	8.33×10^{-4}	3.50×10^{-4}
mixed, equal amounts, antibiotic	0.038	0.610	0.973	8.30×10^{-4}	3.23×10^{-4}
mixed $(0.25, 0.30)$, antibiotic	0.037	0.510	0.973	8.30×10^{-4}	3.22×10^{-4}

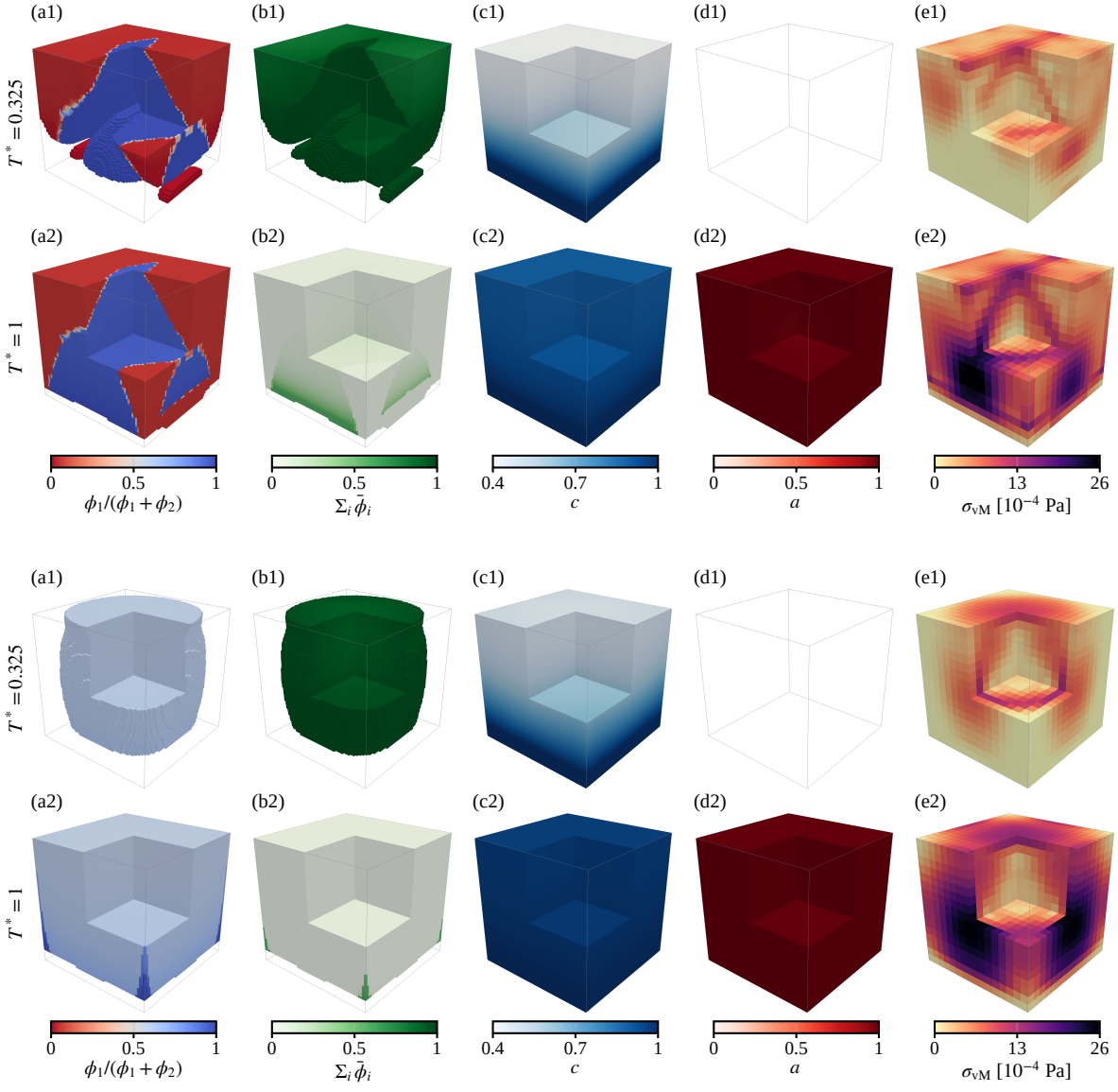


Figure 3: Separate (upper half) and mixed (lower half) example with antibiotic, Python, 64^3, one octant removed, at $T^* = 0.325$ (first row of each half) and $T^* = 1$ (second row): (a) share $\phi_1/(\phi_1+\phi_2)$, (b) living volume fraction $\sum_i \bar{\phi}_i$, (c) nutrient c, (d) antibiotic a, (e) von Mises stress. In the mixed example the living volume fraction falls almost everywhere after the antibiotic.

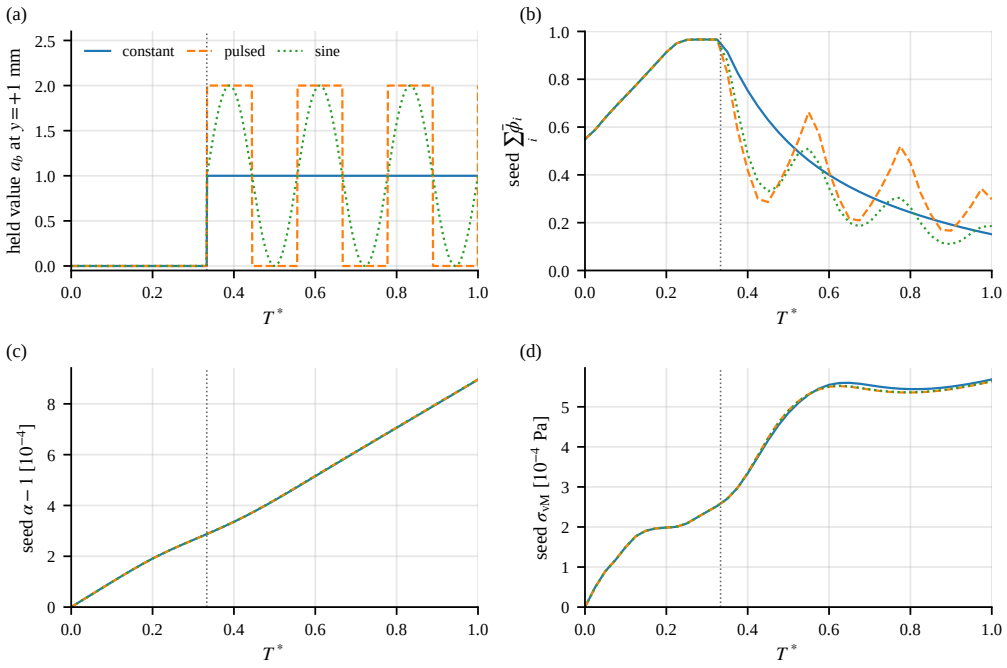


Figure 4: The antibiotic given in three ways with the same total dose from $T^* = 1/3$ (separate example, 64^3): (a) held value at the outer faces, (b) living volume fraction $\sum_i \bar{\phi}_i$ of the seed, (c) seed mean of $\alpha - 1$ and (d) seed mean of the von Mises stress over time for the constant, the pulsed and the sine dose. At $T^* = 1$ the three schedules differ by 0.1 % in $\alpha - 1$ and 0.8 % in the stress.

4.3 Antibiotic over time

With the same total dose given constantly, in pulses or as a sine from $T^* = 1/3$ (Figure 4; separate example, 64^3), $\alpha - 1$ changes by 0.1 % and the seed stress by 0.8 % between the three schedules, while the living volume fraction of the seed follows the dose: in the pauses of the pulsed dose it recovers from 0.17 to up to 0.34 and ends at 0.30, and the sine gives a lower minimum (0.11 against 0.15) and a higher end value (0.19) than the constant dose (0.15). How the dose is given acts on the composition and the living biomass, hardly on the mechanics.

4.4 Which composition does new biofilm receive?

The rule for newly reached points (Section 2.3, item 4) is not fixed by the papers: the point model has no transport term, Eq. (1) moves ϕ_{3D} without carrying the point state, so the composition travels with the biofilm only through the state that a newly reached point is given. The two rules compared here differ in the volume fraction they assign. Inheriting the neighbours' amounts, shared by their living biomass, gives a new point a volume fraction close to that of the established biofilm behind it; the start state of the colony gives it the small start fractions of the colony (0.25 to 0.3 per species). By Eq. (9) the antibiotic term scales as $1/\phi_i^2$, so a point started from the start state loses its living fraction at a rate about ten times that of an inherited point (a/ϕ_1^2 of 16 against about 1.2). Where the biofilm has stopped spreading the rule hardly matters: with the start values of the seed instead of the living composition of the neighbours, the seed values of the cube examples change by a few per cent ($\sum_i \bar{\phi}_i$ 0.147 against 0.152 in the separate example with antibiotic). In the maze of Klempt et al. (2024, Fig. 11), where the biofilm keeps spreading after the antibiotic and most of the colonised area at the end was reached after the onset, the rule decides the outcome (Figure 5, set-up in Section 4.6): the living volume fraction of the colonised region falls to 0.10 shortly after the onset of the antibiotic at $T^* = 20$ and recovers to 0.35 at $T^* = 60$ when new cells take the state of their neighbours, because the biofilm keeps spreading

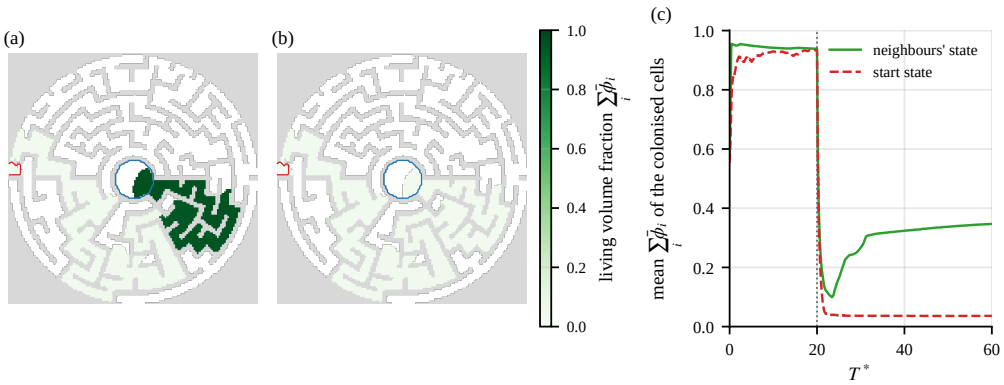


Figure 5: The maze with the antibiotic held at the entrance from $T^* = 20$, at $T^* = 60$: living volume fraction $\sum_i \bar{\phi}_i$ where $\phi_{3D} \geq 0.05$ when newly reached cells (a) take the state of their colonised neighbours and (b) take the start state of the colony; walls grey, nutrient source outlined in blue, antibiotic source in red. (c) Living volume fraction averaged over the colonised cells against T^* for both rules; the antibiotic starts at the dotted line.

with dense living cells that the antibiotic removes slowly, and it stays at 0.04 when they start from the start state of the colony, because thinly started cells die at once. The Galerkin element with the start-state rule gives the same (0.04 to 0.07 in both codes). Without antibiotic the two codes agree in the colonised area within 3 to 5 % and in the living fraction within 0.006.

4.5 What reaches the stress

The growth $\alpha - 1$ hardly depends on the composition, as the bound of Section 2.4 requires: Eq. (3) is unchanged, the antibiotic lowers ψ_i but not ϕ_{3D}, and the seed growth is within 10 % of the bound $k_\alpha T^*$ when the species start apart and within 15 % when they start mixed at $\phi_{3D} = 0.55$ (Table 3: 9.16 against 9.22×10^{-4} in the separate example, about 1 % between the two cases; 8.56 against 8.54×10^{-4} in the mixed one). The composition reaches the stress along the path (6) only, through the front speed, which the nutrient rise from 0.54 to 0.91 raises by 37 % at the edge of the seed, but in these examples the effect of the antibiotic on the converged seed stress is within the numerical uncertainty: 6.55 without and 6.70×10^{-4} Pa with the antibiotic in the separate example (extrapolated, GCI 0.7 and 8.3 %). On a fixed grid the difference is larger (6.73 against 6.00×10^{-4} Pa on 64^3) because the two cases converge from opposite sides. The mixed seed carries about half the stress of the separate one, because its edge is less sharp: the front carries intermediate values of ϕ_{3D}. A limitation follows from the same structure: the front speed $r\,c/(k + c)$ depends on the nutrient only, so dead biofilm keeps spreading after the antibiotic; in the maze about two thirds of the cells reached after the antibiotic hold only dead biofilm when new cells start from the start state. Tying the front speed or the growth rate to the living biomass would open a second path and changes the model (Section 5).

4.6 Geometry: rigid bars and a maze

Klempt et al. (2024) also grow the biofilm against rigid obstacles (their Section 4.3) and through a maze (Section 4.4). Both were repeated with the coupled model in the Python code, without antibiotic unless stated (Figure 6).

Rigid bars. Four bars after Fig. 9 of the paper, with positions read off the sketch: two along x and two along z, each 0.25 mm square, between 0.5 and 0.75 mm from the centre. They hold no biofilm and no nutrient, and the biofilm slides along them without friction. All other settings are those of Section 4.1 (32^3, outer step 0.0025). The biofilm grows around the bars and closes

behind them, and by $T^* = 0.2$ it fills the cube between them. The composition of the seed hardly changes. The bars restrain the swelling: the seed von Mises stress is 7.1×10^{-3} Pa against 7.4×10^{-4} Pa without bars in the separate example, about ten times higher, and 9.5×10^{-4} against 2.1×10^{-4} Pa in the mixed one; the growth $\alpha - 1$ of the seed is 9 and 21 % lower. These are values on one grid; on 32^3 the stress without bars lies 14 % above its extrapolated value (Table 3), small against the factor caused by the bars.

Maze. The maze of Fig. 11 of the paper, read off the figure (560 μm across), in two dimensions with 4 μm cells, the constants of Table 6 of the paper ($\beta = 10$ $\mu\text{m}^2/T^*$, $r = 200$ $\mu\text{m}/T^*$, $k_\alpha = 10^{-6}\,T^{*-1}$), the nutrient held in the centre, the colony of both species at the entrance and the point model of the two-species examples. With the antibiotic, $a = 1$ is held at the entrance from $T^* = 20$. The model is two-dimensional, so it gives no stress. The biofilm follows the corridors, also into dead ends, and approaches the centre: at $T^* = 60$ the nearest point with $\phi_{3D} \geq 0.5$ is 42 μm from the centre without antibiotic and 133 μm with it. The living volume fraction of the colonised region is 0.92 without antibiotic; with it, the rule for new cells decides whether the biofilm recovers (Section 4.4).

4.7 A clinical molar crown

The cube examples place the seed in a box. To show the coupling on a real geometry, the same model and constants were run around the crown of a human molar from the Open-Full-Jaw dataset (Gholamalizadeh et al., 2022) (Figure 7). The crown ($6.7 \times 6.8 \times 5.5$ mm) stands with its cervical base on the bottom face of a 10 mm box and is rigid for the fields (no biofilm and no nutrient inside, zero flux through its surface) and for the stress ($\mathbf{u} = \mathbf{0}$). Nutrient and, from $T^* = 1/3$, antibiotic are held at 1 on the five faces around the crown (the saliva side); the bottom face is zero flux (the root side). The biofilm starts as a cervical ring 0.25 mm thick and 1 mm high, species A on the half $x < 0$ and B on $x > 0$. These choices are assumptions of this work. The grid has 80^3 cells ($h = 0.125$ mm; 40^3 for comparison), the outer step is 0.025, the film is taken as the cells with $\phi_{3D} > 0.5$, and the stress is solved on the field grid with the film surface free.

With the paper's front speed the biofilm spreads along the bottom face and outwards and reaches the side faces by $T^* = 1$. The composition is carried with it: the A and B halves stay separate (film mean share 0.50; 0.956 in the A ring, 0.051 in the B ring). The antibiotic enters with the nutrient ($a = c$ in the film, 0.73 on average, 0.24 at the ring) and acts first on B, as $b_2 = 2$ and Eq. (9) require: the living fraction of the B ring falls from 0.71 at the onset to 0.13 at $T^* = 1$, that of the A ring from 0.81 to 0.57; the film mean is 0.37 (0.95 without antibiotic). Growth and stress hardly depend on the antibiotic, as on the cube (film mean $\alpha - 1$ of 4.3×10^{-4} with and without; von Mises 3.0 and 3.2 mPa, with the largest values in the corners between the film, the crown and the bottom face). Between 40^3 and 80^3 the film means change by 3 % and the film stress by 7 %, most of which is the stress mesh. The example demonstrates the coupling on a real surface; the stress of the cervical region with a growth law is not treated here.

5 Discussion

What the coupling shows. The examples confirm what the structure of the equations prescribes (Section 2.4). The composition reaches the growth field only through the consumption $g\,\phi_{3D}(\sum_i \bar{\phi}_i)\,c$, and from there only through the front speed $r\,c/(k + c)$. When the antibiotic kills the biomass, less nutrient is consumed, the nutrient rises in the seed (its depletion length grows from 1 to 2.6 mm) and the edge moves faster, by 37 % at the nutrient levels of the seed. The effect on the growth is about 1 % and could not exceed about 10 %, because the growth law $\dot{\alpha} = k_\alpha \phi_{3D}$ counts all biomass, living or dead, and the seed is already near the bound $k_\alpha T^*$; the

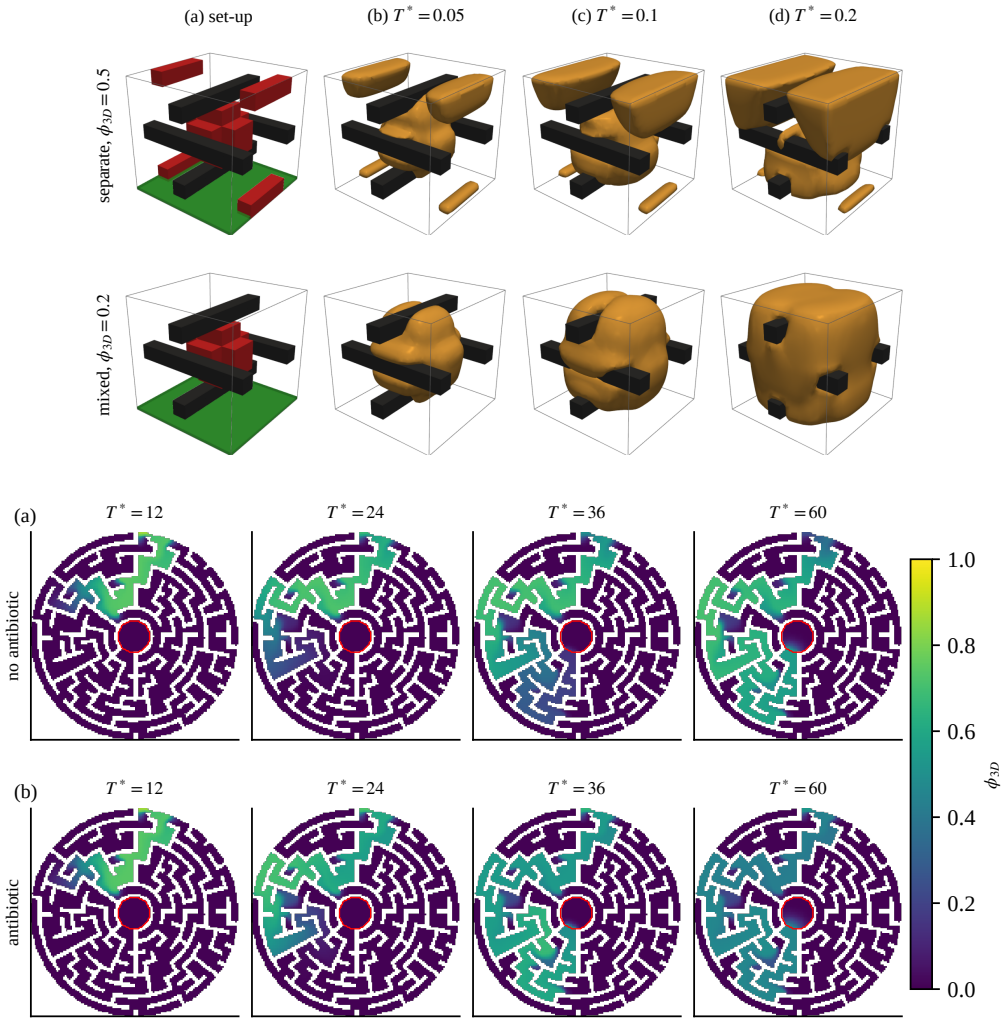


Figure 6: Upper half: growth against four rigid bars (dark grey) after Fig. 9 of Klempt et al. (2024), Python, 32^3, no antibiotic. (a) Set-up: start biomass (red) and nutrient supply on the face $y = -1$ mm (green). (b)–(d) Surface $\phi_{3D} = 0.5$ (separate example) and 0.2 (mixed example, which starts at $\phi_{3D} = 0.55$) at $T^* = 0.05$, 0.1 and 0.2. Lower half: the maze of Fig. 11 of the same paper with the coupled model (two dimensions, 4 μm cells): ϕ_{3D} at $T^* = 12$, 24, 36 and 60, (a) without and (b) with the antibiotic at the entrance from $T^* = 20$. Walls white, nutrient source outlined in red.

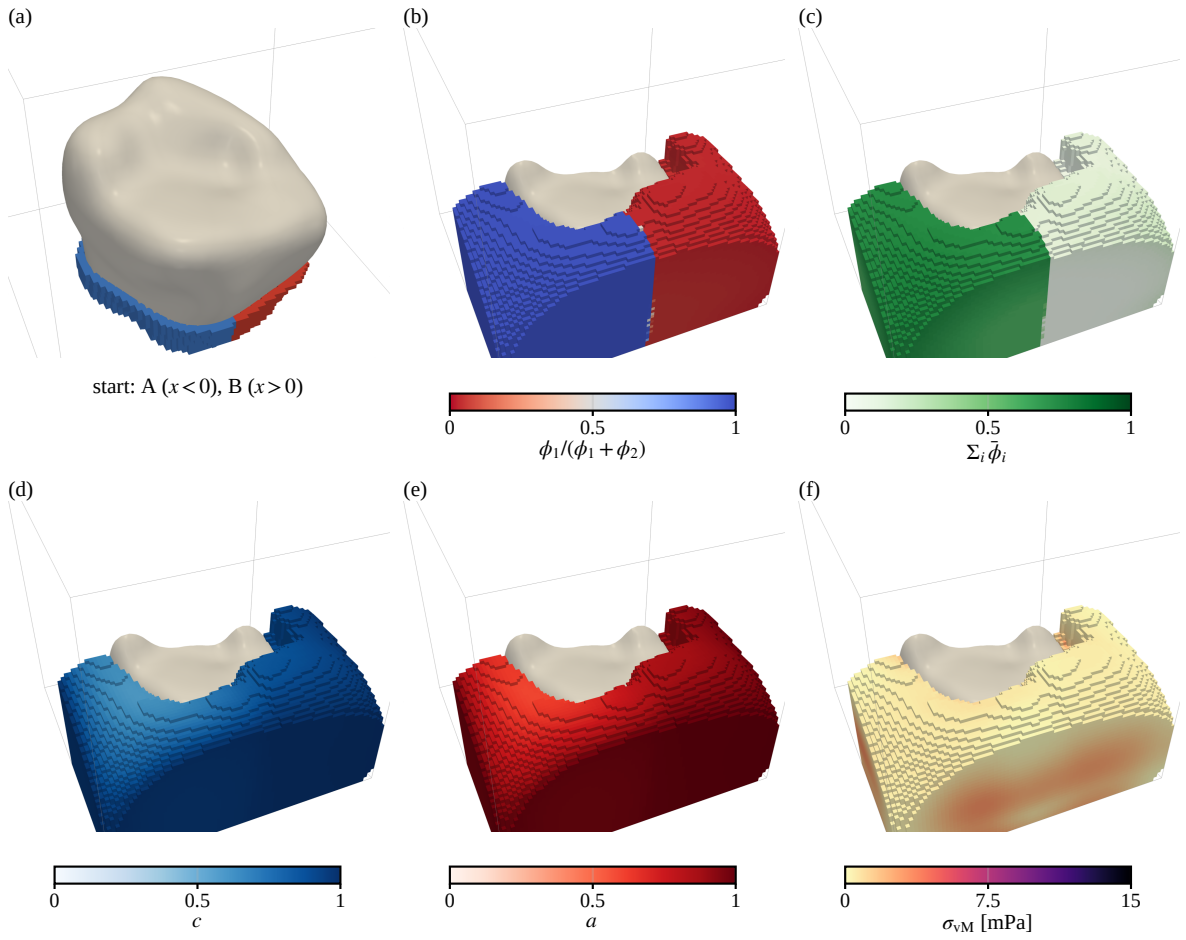


Figure 7: The coupled model around the crown of a human molar (Gholamalizadeh et al., 2022), 80^3 cells, with the antibiotic from $T^* = 1/3$. (a) Start: the cervical ring, species A on the half $x < 0$ (blue), B on $x > 0$ (red). (b)–(f) The film $\phi_{3D} > 0.5$ at $T^* = 1$, coloured by (b) the share $\phi_1/(\phi_1 + \phi_2)$, (c) the living volume fraction $\sum_i \bar{\phi}_i$, (d) the nutrient c, (e) the antibiotic a and (f) the von Mises stress.

effect on the seed stress is within its grid uncertainty, because the stress is set by the jump of α across the edge of the seed and the edge moves but does not change its height. The local nutrient decides where the antibiotic acts, and the balance of Eq. (9) says how: a species declines at a rate $\alpha^* b_i/(\eta_i \phi_i^2) - c^* A_{ii}/\eta_i$, slow where the nutrient is high and the species is dense, fast where the nutrient is low or the species is thinly present. The same dose therefore removes a species in one part of the biofilm and not in another, which a model without space cannot show, and removes it ten times faster from a mixed seed than from a seed of one species, which a model without the point model's balance cannot show either. Numerically, the stress depends on how sharp the edge of the biofilm is, and that edge is set by the front term, so the coupling needs a second-order treatment of that term (Section 3).

Limitations. (i) The front speed depends on the nutrient only, so dead biofilm keeps spreading after the antibiotic. Tying the speed to the living biomass is a change of the model; in a test on 16^3 (front speed multiplied by the local living fraction) the seed stress rose by 51 % in the separate example and about fourfold in the mixed one, so this choice matters for the stress. (ii) Eq. (3) lets dead biomass grow at the same rate as living biomass. Growth from the living biomass, $\dot{\alpha} = k_\alpha \phi_{3D} \sum_i \bar{\phi}_i$ as in Wanner and Gujer (1986), would make the antibiotic act on the growth directly; its size can be read from the time series of the seed, since $\alpha - 1$ would then scale with the time average of $\sum_i \bar{\phi}_i$: 0.54 against 0.92 with and without the antibiotic in the separate example and 0.30 against 0.88 in the mixed one, a reduction of the seed growth by about 40 and 65 % in place of the 1 % of the published law. (iii) The point model has no space; the composition moves only through the rule for newly reached points, which decides in the maze whether the biofilm recovers. A species-resolved continuum would replace the rule. (iv) The mechanics is coupled one way, as in Klempt et al. (2024); with $E = 10$ Pa the neglected term acts only for coefficients far beyond the paper values (an effect on the growth from about 10^5 on, 16^3 test). (v) Living and dead biomass, and both species, have the same stiffness; a species-dependent stiffness changes the seed stress by 9 to 89 % in a 16^3 test, depending on which species is stiffer. (vi) The examples use synthetic data with known parameters, the stress converges slowly when the species start mixed, and the element's values carry a grid error of 16.5 %. Calibrated data (for example the five-species data of Heine et al., 2025) and a measured stress are needed before the stresses can be validated, not only verified.

Outlook. Species-resolved transport, so that the composition moves with the biomass; front speed and growth tied to the living biomass, with the consequences for the stress quantified above; growth laws with a mechanical feedback; calibrated multi-species data in the element; and a realistic tooth or implant geometry. Each can be added to the coupling as presented here, and the verification procedure, several codes, a manufactured solution and grid convergence indices, carries over.

6 Conclusions

This work coupled the multi-species point model of Klempt et al. (2026) to the continuum growth model of Klempt et al. (2024) at every integration point, without changing the equations of either model. The point model receives the local nutrient and antibiotic from the fields, $c^* = c_0^* c$ and $\alpha^* = \alpha_0^* a$, and its living volume fraction sets the uptake $g\, \phi_{3D}(\sum_i \bar{\phi}_i)\, c$ of both.

1. The structure of the coupled equations fixes what the composition can change. It reaches ϕ_{3D}, α and the stress only through the nutrient and the front speed $r\, c/(k + c)$, because the growth law $\dot{\alpha} = k_\alpha \phi_{3D}$ counts all biomass; the seed growth is bounded by $k_\alpha T^*$ and lies within 10 % of it, and the front speed changes by at most a factor of two. The quasi-steady nutrient depends on the composition only through $\phi_{3D} \sum_i \bar{\phi}_i$ and, with the Thiele modulus

$\Lambda^2 = g\ell^2/d = 1$, is depleted over the half edge of the domain but never exhausted. The time link s is equivalent to the viscosities η_i/s and is fixed by the data.

2. In the point model the antibiotic acts through a balance, $c^* A_{ii} \phi_i^2$ against $\alpha^* b_i$, whose decay rate grows as $1/\phi_i^2$: a species that fills its point is removed slowly, the same species thinly present at once. This explains three results of the examples: the local nutrient decides where the antibiotic acts; a mixed seed loses its living biomass ten times faster than a seed of one species (0.96 to 0.04 against 0.97 to 0.15); and the state given to newly reached material, which neither paper fixes, decides in a maze whether the biofilm recovers after the antibiotic (living volume fraction 0.35 against 0.04).

3. With the published equations the antibiotic changes the growth by about 1 % and the converged seed stress within its grid uncertainty, in the cube and around a molar crown. Growth from the living biomass would lower the seed growth by about 40 to 65 % in the same runs, and a front speed tied to the living biomass raises the seed stress by 51 % to fourfold; both are changes of the model.

4. The coupling was implemented in a finite-volume code, in a neighboured-element user element in ANSYS and in a Galerkin user element in Abaqus, and verified by replay of the point model, a manufactured solution and grid convergence indices. The front term $-r \frac{c}{k+c} \mathbf{n}_c \cdot \nabla \phi_{3D}$ sets the accuracy of the stress: a constant added diffusion gives a quarter of the converged seed stress, a second-order treatment is needed; with it the growth converges with a GCI of 0.7 to 3.3 % and the seed stress with 0.7 to 8.3 % when the species start apart.

The next steps are models in which the front speed, the growth or the stiffness depend on the living biomass or on the species, with the consequences estimated here.

Data availability

The finite-volume code, the point-model routine, the run scripts, the result files of the examples and the figure scripts are in the public repository `https://github.com/keisuke58/pde-fem-biofilm`. The examples use the published synthetic data of Fritsch et al. (2025); no calibration data are used in this work. The molar crown of Section 4.7 is derived from the Open-Full-Jaw dataset of Gholamalizadeh et al. (2022), licensed CC BY-NC-SA, and the derived surface is redistributed in the repository under the same licence. The user elements depend on the commercial codes and on the element of Rudolf et al. (2025); the material routines are in the repository.

TODO: Confirm with all authors which parts of the element code may be published, and whether a release (DOI) of the repository is made at submission.

Declarations

Author list, contributions, funding and conflicts of interest: to be agreed.

TODO: Authors, affiliations, contributions, funding (LUH/IKM, Keio), acknowledgements; apply the Springer Nature template (sn-jnl.cls) of Biomechanics and Modeling in Mechanobiology at submission.

A Numerical details

A.1 The neighboured element

Three details of the element were corrected, without changing its equations: the surface points take the value of the nearest Gauss point instead of a distance-weighted mean, the Laplacian is

used in a conservative form, and the neighbours of the fit are chosen by distance instead of by element number. Before, ϕ_{3D} leaked through the walls of the cube (the surface points took an inverse-square mean, and the column sums of the Laplacian were not zero), and the stencils of the 16^3 and 24^3 grids were asymmetric inside the domain. With a manufactured solution the corrected element still does not converge from 8^3 to 16^3 (L_2 error of ϕ_{3D} 2.2×10^{-3} to 1.8×10^{-3}, of c 8.2×10^{-3} to 1.2×10^{-2}): the MLS stencils of the Gauss points next to a face include the surface points and are ill-conditioned (error of the Laplacian of the manufactured c: 0.88 on 8^3, 65 on 16^3, against $|\nabla^2 c| = 3.5$). Rebuilding these stencils with the neighbours mirrored across the faces (zero flux built in) makes the operators consistent up to the walls, and with the weight width proportional to h the element converges (ϕ_{3D} 9.5×10^{-4} to 3.4×10^{-4}, order 1.5; c 9.0×10^{-4} to 2.2×10^{-4}, order 2.0). In the coupled example the mirrored operators run on 8^3 but become unstable at a corner of the 16^3 grid after the antibiotic onset; they are not used for the results. The explicit update of the element is limited by $\beta \Delta t / h^2 \leq 0.12$ to 0.16 on 16^3 and 24^3 (Bloch analysis of the stencil); the Lax–Wendroff form with the extremum sensor keeps ϕ_{3D} in $[-0.03, 1.01]$ on 64^3 in Python but is unstable at the corners of the supply face on 24^3 in the element, so the element's stresses are quoted from the upwind MLS gradients. The time-step series of the element and the stability analysis will be given in the supplementary material.

B Verification details

B.1 Manufactured solution and mechanics

Sources for ϕ_{3D} and c were built from a manufactured solution (Roache, 2002) by automatic differentiation and checked symbolically (agreement 10^{-14}). The finite-volume and the Galerkin code converge at second order in L_2 (the error plot is in the supplementary material): observed orders between 8^3, 16^3 and 32^3 of 1.85 and 1.90 for ϕ_{3D} and 2.03 and 2.01 for c in Python, 1.68 and 1.81 for ϕ_{3D} and 2.00 and 2.05 for c in Abaqus; on 32^3 the errors of ϕ_{3D} are 2.0×10^{-4} and 2.9×10^{-4}. For the SUPG form the order of ϕ_{3D} is about 1.5: the largest error sits in a thin layer next to the wall opposite the supply, where $\nabla c \to \mathbf{0}$ and the front direction changes over a few cells. The analytic tangent of the Abaqus element agrees with central differences to 5×10^{-11}. The last growth field of an Abaqus run, solved by the Python mechanics, gives the seed von Mises stress to 1.1 % (no obstacles) and 0.5 % (obstacles with frictionless contact).

B.2 The front term and the agreement of the codes

Four treatments of the front term were compared, none of which changes Eqs. (1) to (3): (1) a constant added diffusion $rh/4$ (first order; the element as delivered needs it to stay stable); (2) TVD upwinding (MUSCL with the minmod limiter; Python); (3) SUPG with $\tau = \frac{h}{2|v|}(\coth Pe - 1/Pe)$ on the strong residual (Abaqus); (4) for the explicit element, either the Lax–Wendroff term $\frac{\Delta t}{2}\, v \cdot \nabla(v \cdot \nabla \phi_{3D})$ with an added diffusion switched on by an extremum sensor, or upwind-biased MLS gradients. Figure 8 shows the seed means of $\alpha - 1$ and of the von Mises stress at $T^* = 1$ of the separate example with antibiotic against the grid for every scheme and code. The growth variable converges for every scheme, and the codes agree within their grid-convergence bands: Python extrapolates to 9.0 to 9.8×10^{-4}, Abaqus to 8.8 and 9.45×10^{-4}, the element to 8.7×10^{-4} with upwind gradients. The stress does not converge with the constant added diffusion: on 16^3 it is about a quarter of the converged value (1.3 to 1.7 against 5.9 to 6.3×10^{-4} Pa from the two second-order schemes of two codes), and on 96^3 it is still rising. Treatments 2 to 4 give the same stress within a few per cent from 32^3 on (Python TVD 5.90 and 5.98×10^{-4} Pa on 32^3 and 64^3, Abaqus SUPG 6.27 and 6.22×10^{-4} Pa on 32^3 and 48^3, GCI 1.2 %). The element with upwind gradients gives 2.14, 4.56 and 5.15×10^{-4} Pa on 8^3, 16^3 and 24^3, extrapolated 5.8×10^{-4} Pa (GCI 16.5 %), within the band of the second-order schemes; its MLS operators over 30 neighbours give it a resolution of about one element, not one Gauss-point

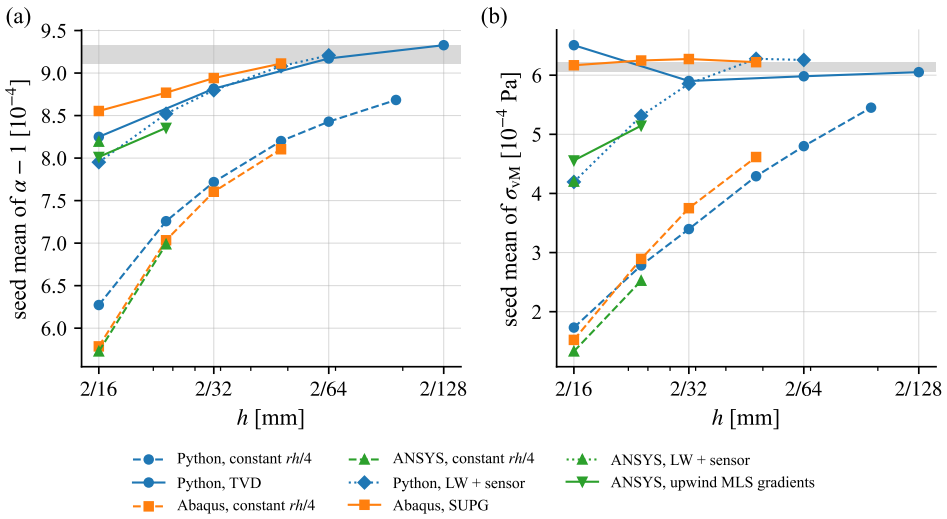


Figure 8: Separate example with antibiotic: seed means of (a) $\alpha - 1$ and (b) the von Mises stress at $T^* = 1$ against the grid, one line per scheme and code (dashed: constant added diffusion $rh/4$; solid and dotted: the second-order and the stabilised explicit forms). Grey band: the finest grids of the two second-order schemes. These runs start newly reached points from the seed composition (the alternative rule of Section 4.4) at a field step of 0.00125.

spacing (with the same added diffusion, the element on 16^3 and Python on 32^3 differ by 1 % in $\alpha - 1$ and 5 % in the stress).

B.3 Time step, regularisation and stability

In the element codes the point model advances in every increment (0.00125 in T^*; Abaqus: observed order 1.1 in time, error against the extrapolated value 0.8 % in $\alpha - 1$ and 2.2 % in the stress). In the Python code the point model and $\sum_i \bar{\phi}_i$ are updated once per outer step ΔT^*, inside which ϕ_{3D}, α and the fields take explicit RK4 steps. The composition and $\alpha - 1$ change by less than 3 and 0.3 % between outer steps of 0.02 and 0.0025; the stress does not: on 64^3 the separate example with antibiotic gives 5.68, 5.82, 5.94 and 6.00×10^{-4} Pa at $\Delta T^* = 0.025$, 0.01, 0.005 and 0.0025 (first-order extrapolation 6.07), the mixed one 3.23, 3.09, 2.90 and 2.82×10^{-4} Pa (2.74). The sensitivity comes from the front speed: after the antibiotic onset $\sum_i \bar{\phi}_i$ is small, ∇c is weak and the front direction responds to small changes of the consumption. Within an outer step, the inner RK4 step, a trapezoidal update of $\sum_i \bar{\phi}_i$ over the step and letting newly reached points consume at once each change the stress by less than 1 %; the one-increment lag of $\sum_i \bar{\phi}_i$ and the regularisation of the front direction ($\varepsilon^2 = 10^{-6}$ against 10^{-8}, 0.1 %) are not the cause. A larger regularisation ($\varepsilon^2 = 10^{-4}$) changes the mixed stress by 30 %, which shows that the front direction, not the speed, carries the sensitivity. Without antibiotic the outer step changes the stress by 2.5 % (separate) and 12 % (mixed). The grid series of the examples are therefore run at $\Delta T^* = 0.0025$. The explicit inner step is limited by the diffusion number $D = \beta \Delta t / h^2 \leq 0.15$ and the Courant number of the front $Co \leq 0.4$; on 192^3 both limits are close at the same time and the step diverged in the first outer step, which the combined condition $12D + 2Co \leq 2.2$ removes without changing the step on 32^3 to 128^3.

Table 3: Grid convergence (Python, TVD, outer step 0.0025): seed means at $T^* = 1$, observed order p, extrapolated value and GCI (Celik et al., 2008). Mixed: equal amounts unless the start fractions (ϕ_1, ϕ_2) are given; antib.: with antibiotic; –: not yet in the asymptotic range.

example	quantity	32^3	64^3	128^3	p	extrapolated	GCI
separate, antib.	$\alpha - 1$ [10^{-4}]	8.453	8.931	9.112	1.4	9.22	1.5 %
separate, antib.	von Mises [10^{-4} Pa]	5.523	6.000	6.284	0.75	6.70	8.3 %
separate	$\alpha - 1$ [10^{-4}]	8.729	9.014	9.110	1.6	9.16	0.7 %
separate	von Mises [10^{-4} Pa]	7.440	6.730	6.585	2.3	6.55	0.7 %
mixed, antib.	$\alpha - 1$ [10^{-4}]	7.42	8.04	8.32	1.2	8.54	3.3 %
mixed, antib.	von Mises [10^{-4} Pa]	2.00	2.82	3.30	0.76	3.99	26 %
mixed	$\alpha - 1$ [10^{-4}]	7.46	8.08	8.35	1.2	8.56	3.1 %
mixed	von Mises [10^{-4} Pa]	2.13	3.08	3.65	0.74	4.50	29 %
mixed (0.25, 0.30), antib.	$\alpha - 1$ [10^{-4}]	7.42	8.04	8.31	1.2	8.51	3.1 %
mixed (0.25, 0.30), antib.	von Mises [10^{-4} Pa]	2.01	2.96	4.01	–	–	–

B.4 Grid convergence table

Supplementary material

The L_2 error plot of the manufactured solution, the discretisation table of the three codes, the time-step series of the neighboured element and of the Galerkin code, the stability analysis of the explicit element (Bloch analysis of the stencil, step limits $\beta\Delta t/h^2 \leq 0.12$ to 0.16), the sensitivities to $\phi_{\min}$, d, ε, f and the start rule, the test of the mirrored operators of the element, and the animation of the maze will be provided as supplementary material.

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